

Building a unified science of sensorimotor control with Densely Observable Measurement Environments [DOMEs]

FreeMoCap Foundation, Inc.

Written Proposal to the NSF X-Labs Initiative

1. Mission

The MISSION of the **FreeMoCap Foundation X-Lab (FMC-X)** is to develop and disseminate the tools to measure a new scientific observable representing the closed loop of the **perceptual, mechanical, and environmental** aspects of goal-directed behavior of freely moving organisms to reshape perceptual/motor neuroscience, musculoskeletal biomechanics, embodied/agentive artificial intelligence, and mobile robotics into a new convergent science of **sensorimotor control in natural environments**.

1.1. Measurement target

We will organize our technical development around the technical bottlenecks which limit our ability to combine reconstructions the **eye, body, and environment** into combined representations which trace to activity in the central and peripheral nervous system ...

- Best Eye trackers ever made at any price
- Best Mocap system ever made at any price
 - (but we'll sell em **real** cheap)
- Completely new fully integrated platform, exposing entirely new observable defined by densely overlapping measurements from the DOME
- DOMEs **everywhere**
 - Every human or animal-focused lab
 - Every hospital and PT clinic
 - DOMES on the space station and Vomit Comet rocket (study sensorimotor control and VOR in microgravity)
 - DOMES in athletic centers - Training athletes for high performance through education into their underlying sensorimotor control systems
 - DOMES in mobile robot and drone labs - Align with human research, stop the robot folks from handrolling their own mocap all the time
 - DOMES at the bottom of swimming pools and integrated into SCUBA systems of underwater welders (and other specialized trades)
 - Mobile DOMES studying animals in natural environment, mounted to the bottoms of ships and integrated with ultrasonic sensors to mocap dolphins playing in the wake
 - Tiny DOMES to study honeybee and dragonfly visual flight control
 - DOMES in every classroom!
 - In both science/tech/health classrooms AND animation/game design classrooms (DOMEs for animation mocap, freemocap already proved this works)
 - Always build "cheaper than a textbook" version of all hardware
 - Commensurate data output (with less precision and fewer modalities)
 - Prepare new generation for a future where human and robot centered research is focused on full-body kinematics and 3d binocular eye tracking with retinal projection.
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1.2. Flagship

We will develop and validate a **novel research instrument** we call a **DOME** (*Densely Overlapping Measurement Environment, Ref - Figure 1 - Hero Figure showing full DOME*), which unifies a heterogeneous array of multi-modal sensors into single, centrally-controllable, metrologically-validated instrument for the measurement and manipulation of a freely moving agent within a specified 3D volume of real-world space (ref DOME-L, DOME-S, and DOME-Mobile in 3. Outcomes).



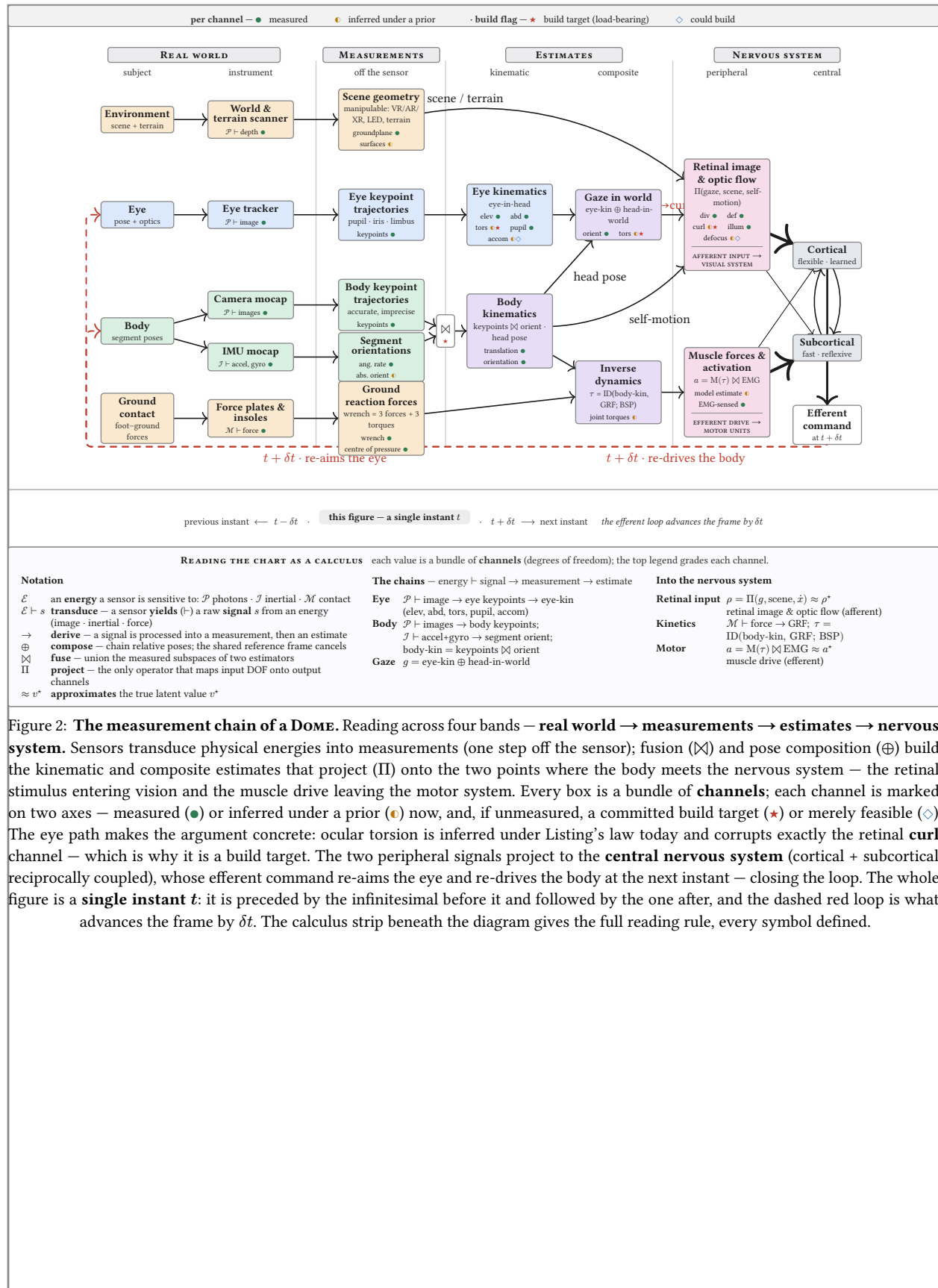


Figure 2: **The measurement chain of a DOME.** Reading across four bands — **real world** → **measurements** → **estimates** → **nervous system**. Sensors transduce physical energies into measurements (one step off the sensor); fusion ($\bowtie$) and pose composition ($\oplus$) build the kinematic and composite estimates that project (II) onto the two points where the body meets the nervous system — the retinal stimulus entering vision and the muscle drive leaving the motor system. Every box is a bundle of **channels**; each channel is marked on two axes — measured ($\bullet$) or inferred under a prior ($\circ$) now, and, if unmeasured, a committed build target ($\star$) or merely feasible ($\diamond$). The eye path makes the argument concrete: ocular torsion is inferred under Listing’s law today and corrupts exactly the retinal **curl** channel — which is why it is a build target. The two peripheral signals project to the **central nervous system** (cortical + subcortical, reciprocally coupled), whose efferent command re-aims the eye and re-drives the body at the next instant — closing the loop. The whole figure is a **single instant t** : it is preceded by the infinitesimal before it and followed by the one after, and the dashed red loop is what advances the frame by δt . The calculus strip beneath the diagram gives the full reading rule, every symbol defined.

2. Technology Landscape

- Existing tools and methods
 - Some neat softwares and CV/ML tools, but all single threads, no unified tool
 - OpenSim - Widely used, very difficult to operate, highly specialized niche within biomechanics, single-modality
 - DeepLabCut - Widely used, very difficult to operate, single-modality
 - OpenCap works, but only for iPhones and requires a cloud upload, and caters exclusively to clinical biomechanics. Single-modality, non-convivial.
 - SIMPL/Meshcapade - Has built past empirically validity. 3d from 2d and mesh-fitting pipelines not truth preserving. Building increasingly more abstract and less tracable data channels on top of the same 2d data we've had for a while
 - etc
 - MuJoCo/IsaacLab- Driving revolution in mobile robotics! We will directly contribute to this by feeding RL robots DOME data, and then using resulting control theory to design controlled experiments (see ARGPv3)
- FreeMoCap and its landscape
 - Established Tool with Growing International Community
 - >15k uses from 152 countries, 10k GH stars, 3.5k in Discord server
 - Existence is proof-of-concept of approach, XLabs scale support would accelerate FreeMoCap model into the global scale
 - v2 has realtime and professional architecture (thanks to EI)
 - DOME-L and DOME-S camera-based mocap will use FMC or a variant. Mocap, Eye, World camera software composable from FMC polyrepo (skellycam, skellytracker, skellyforge, etc)
- ARGP
 - ARGPv1 - PhD Era¹⁻⁴
 - ARGPv2 - NEU Era (Fuzzed out due to complexity, frustration spawned freemocap)
 - DOME includes ARGPv3
- Laser Skeleton research
 - ⁵⁻⁷
 - etc
 - Ferret work
 -
- Existing Instruments
 - **Modern mobile eye trackers suck**
 - Pupils labs - Used to be cool, black box proprietary now. Best Available, used in Ferret and Laser Skeleton work.
 - Tobii et al not worth mentioning
 - Missing DoF's!
 - Can't measure Torsion
 - Field has invoked Listing's Law to ignore torsion for decades
 - Listing's Law is NOT APPLICABLE when VOR is active
 - VOR is active during 100% of natural behavior

- ▶ Torsion directly affects axis that Retinal Curl⁶ lives on, so inability to measure it corrupts signal we want to propagate into visual cortex
- ▶ We know how to measure Torsion (iris texture), and Lens Accommodation too, while we're at it (yates et al DPI)

- ▶ **Modern mocap sucks**

- Old, awful, expensive, black boxes
- So difficult to use it's a literal bottleneck
- Difficult limits lab size (Can't re-target cameras without shutting down the lab for a week. Can't maintain multiple configurations)
- Theia exists, but it's re-building the same problems of the previous generation in a new modality

3. Outcomes

- Develop, validate, and disseminate - DOME Variants
 - **DOME-L** — the flagship instrumented volume, in the greater Boston area. Large enough to fully enclose the smaller variants for validation and metrological grounding.
 - **DOME-S** — the extension of the commodity-camera capture volumes already built and disseminated through the FreeMoCap Project. Representative of the DOME a research lab or a classroom would build.
 - **DOME-Mobile** — a wearable, self-contained DOME producing Dense Observations in unconstrained indoor and outdoor environments, continuing the lineage of the PI's gaze/gait and retinal optic flow work⁵⁻⁸.
 - Also - disseminate custom use-directed builds of DOME variants through network of collaborators studying Humans, animals, and robots.
- New eye tracker
 - Best ever made
 - Add every sensor we can, see which allow us to measure DoFs we need, optionally scale back and remove
 - Optionally also make cheaper version for classrooms - fully compatible data output (with less precision and depth)
 - Remake the field by retraining the next generation of researcher in the tools of the field you want to build
 - Dense instrumentation in the eye and world cameras
 - **Eye sensor array**
 - includes
 - High res IRGB
 - High speed IRGB
 - MEMS single-pixel sensor
 - LiDAR scanner
 - IR/Structured IR illuminators
 - Event Camera
 - **World sensor array**
 - Build different form factors (big heavy, small light)
 - Head- or Drone- mountable
 - Stereo RGB, Structured IR, LiDAR, etc
 - High density multi-model environment scan
 - Fast scan (RGB-D)
 - Slow scan (Textured Mesh)
 - Semantic segmentation (via SAM3, YOLO, etc)

3.1. Timeline		
Phase	Name	Description
1	Organize	Set up Organization (Phase 0) – Solidify release of FreeMoCap v2
2	Architect	Develop core software architecture, define key ontologies and entity/action pairings via use-oriented development (ferret, mouse, guinea fowl, marmoset, human research). Design and prototype instrumentation (mocap/eye cameras, electrode arrays and miniscopes, motor-unit EMG, environmental control systems)
3	Execute	Integrate new architecture into global/public facing community of users (presented as FreeMoCap v3). Iterative design of instrumented system
4	Stabilize	Establish OODA loop within expected standard operations
5	Expand	Once stabilized, expand key performance metrics
6	Align	Focus on building deeply integrated community of users and contributors (Transition BDFL → BDFaSPoT)
7	Establish	Focus on long term stability and healthy growth
8	Maintain	Beyond proposal timeline – Establish organization practice in service of unbounded infinite timeline. Apply winning model to support extending ‘mesoscale research revolution’ to all biological sciences (e.g. integrating deeper microbio/biochem insights beyond the scope of the present perceptuomotor/biomech/robotics focus)

Table 1: Timeline of Phased Milestones

4. Senior/Key Personnel Qualifications

4.1. Jonathan Matthis, PhD [Principle Investigator]

4.2. EI

4.3. AC,PhD

4.4. RR, CPA or w/e

Name / Role	Qualifications
Jonathan Matthis, PhD President	• Founder and chief maintainer of FreeMoCap project • Left tenure track academic research position precisely because the institution could not support the work that needed to be done • Decades experience integrating visual neuroscience with biomechanics/robotics⁵⁻⁷
NR, CEO	• Expertise in technically and software dev • Previous CEO XP, founded and ran own buisness • CEO helps PI apply his specific expertise to the core tasks of the XLab
EI CTO	• Decades of experience in software Industry • Chief architect and CTO for established companies • Represents expertise truly absent from academic contexts • Ensures world-class enterprise-level software capability • Will build training/workshop/hackathon backend to inject EI-level software development expertise into broad community of science
AC, PhD CSO	• Expertise in validation of clinical research tools and bench-to-bedside development • [CITE DISSERTATION] • Experience in neuroprosthetics design and brain imaging • Worked with PI on core FMC dev and validation
JKL, PhD CAIO	• PhD - Cognitive Science • Director - Cognitive Science Program • Current - Senior AI Applications Engineer @ SOLID • Expertise in use of AI in classroom, lab, and enterprise
RR CFO	• Bookkeeping • NSF Grant management • Finances • Entrepreneurship (founded own Bookkeeping coop) • Taught Finance Theory at university level
KM Project Manager (?)	• Key driver of⁷ • Worked with PI on core tech⁶ • Expertise in key technologies
MN Project Manager (?)	• Expertise in clinical biomechanics, • Medical systems design, • Mechatronics for clinical tools, • Multi-modal motion capture lab management

Table 2: Senior/Key Personnel Qualifications

5. Team Capabilities Statement

5.1. Senior Team Capabilities

- Jon - done it before in humans, currently doing it in ferrets and mice. Left tenure track position to run FreeMoCap, XLab funding will accelerate that mission
- EI - Deep professional experience in Tech Sector, high level software architecture experienced necessary to develop truly global scale enterprise level of software in open-source/academic contexts
- RR - Organizational health and entrepreneurial development
- NR - XP as CEO and working in tech industry. Hiring her as CEO allows Jon/PI to focus on applying their domain expertise on the core work of the lab
- JKL - AI expertise in teaching, research, and industry. Will build internal AI systems for core software, developer, user, and student assistance.
- AC - focusing clinical adoption and bench-to-bedside transitions. Wrote his dissertation on validation of freemocap. Has all skillsets needed for DOME-S
- MN - XP in mechatronics and large-scale lab building and management. Has core skillset needed for DOME-L
- KM - Part of original laser skeleton team, has all necessary sub-skills to build core instrumentation of Mobile DOME

5.2. Collaborators

Initials	Model / Domain	Expertise
BS	Ferrets	Visual/Perceptual neuroscience, ephys
MD	Guinea Fowl	Musculoskeletal Biomechanics, muscle units
DF	Mice	System Biology
AH	Marmosets / NHP	Ephys
MH	Human	Vision and CPS
Mayo Clinic	Human	Brain scans and implanted sensors
BD, GN/RAI, CH, JH	Robots / Fabrication	Control theory
AA	Physical facilities	Rapid iteration, recruiting

Table 3: Collaborator Network

5.3. Governance

- FMC-X will exist as a independent project within FMC-F organization
- FMC-F will handle other work (e.g. FreeMoCap Project), but otherwise the missions almost fully overlap
- Nearly identical leadership between FMC-F and FMC-X, with final say on all topics falling to the President (who is also the PI) ensures fast decision making (no higher management to talk to - PI is the top of the org chart)
- Collaborator network and user community consulted for design decisions, work targets, and roadmap
 - Handled via “Request for Comments” style announcements (like Python’s PEP), so progress can’t be stalled by community problems.
- Annual Workshop/Hackathon/Conference/Congress to bring together collaborator network to discuss progress, tech targets, friction, etc.

- Workshops to train students and out-of-domain experts
- Hackathons to on-board developers and foster in-person connections between normally remote workers (important for FOSS)
- Conferences to discuss progress and research done with DOME tech
- Congress to discuss org status, plans, and future (annual State of the Skelly address)
- Developer fund, Community Grants Program, etc to support FOSS network
 - Use Community Grants Program to test different strategies for community engagement and consensus taking (Eurovision style voting, DAO methods, etc)

5.4. Autonomy

- We are already autonomous, as of 2 weeks prior to the deadline of this proposal.
 - This project will happen no matter what, but with XLabs funding, we could truly change the world

Bibliography

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